



Evaluating NYC DOT Response Efficiency Using 311 Complaint Data

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NYC
311

Agenda

1. Introduction
2. Context
3. Methodology
4. Findings
5. Conclusion

Our Team - C633 Pod B



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Introduction : 311 Service Requests to the DOT

311 service requests allow NYC residents to report various issues they encounter in the city.

We are analyzing **NYC 311 service request data** filtered to the Department of Transportation (DOT) complaints to **evaluate how efficiently the city responds to infrastructure-related issues.**

We are presenting to DOT leadership, city operations teams, and policymakers, as well as residents who are impacted by service response times.

Key Questions

1. How efficient is the NYC Department of Transportation in resolving 311 complaints across boroughs?
2. What areas of the city generate the most complaints?
3. Which complaint types take the longest and shortest to resolve?
4. What are the most common complaint types?
5. How does response efficiency vary by location?
6. How do complaints trend over time?

Executive Summary

Borough differences in efficiency are moderate, but Queens consistently underperforms, with the highest average resolution time (~12 days) compared to Manhattan (~9.5 days), indicating slower service efficiency.

Resolution time variation is strongly driven by complaint type, with infrastructure issues (e.g., Wayfinding, Bike Rack Condition, Missing Street Signs) taking **30+ days on average**, while Building Conditions and Public Toilet issues are resolved much faster.

Queens and Brooklyn handle the highest complaint volumes, especially for street-related issues (Street Condition, Traffic Signal, Sidewalk), suggesting heavier operational burden in these boroughs.

Despite volume differences, overall resolution times across boroughs are relatively similar (9.5–12 days), indicating system-wide consistency but localized inefficiencies in specific borough-complaint combinations.

Context

What is 311 Service Request ?

311 service requests system connects residents to their local government services for non-emergency issues.

- Started as a hotline service but is now a **consolidated multi-channel system** offering communications via a website portal, social media, and mobile app.
- This makes it very **accessible for residents to report issues anytime and anywhere**. It allows cities to **collect accurate real time data** and sort them by departments and categories to have a **systematic approach to serving the city**.
- This system has the potential to improve resident experience by enhancing **government-to-resident communication** and **service efficiency**.

Context (cont.)

What is NYC DOT ?

The New York City Department of Transportation (NYC DOT) is the city agency responsible for planning, building, and maintaining NYC's transportation system.

DOT manages:

- 6,300 miles of streets and highways
- over 12,000 miles of sidewalk
- approximately 800 bridges and tunnels
- over 1 million street signs
- 13,500 signalized intersections
- nearly 400,000 streetlights
- over 350 million linear feet of markings
- 15,000 parking meters and 37 parking facilities
- Staten Island Ferry, serving 16 million annual riders

Methodology

1. Data Collection

- Sourced **NYC 311 complaint** data from **NYC Open Data**
- Focused on **DOT**-related service requests between **2024 - 2025**

2. Data Cleaning

- Used **Excel / Google Sheets / Python**
- Removed null values and incomplete records
- Filtered for valid, closed cases only

3. Exploratory Data Analysis (EDA)

- Analyzed complaint trends, resolution times, and distributions
- Compared metrics across boroughs and problem types

4. Data Visualization

- Built interactive dashboards in **Tableau**
- Highlighted key patterns, outliers, and performance differences

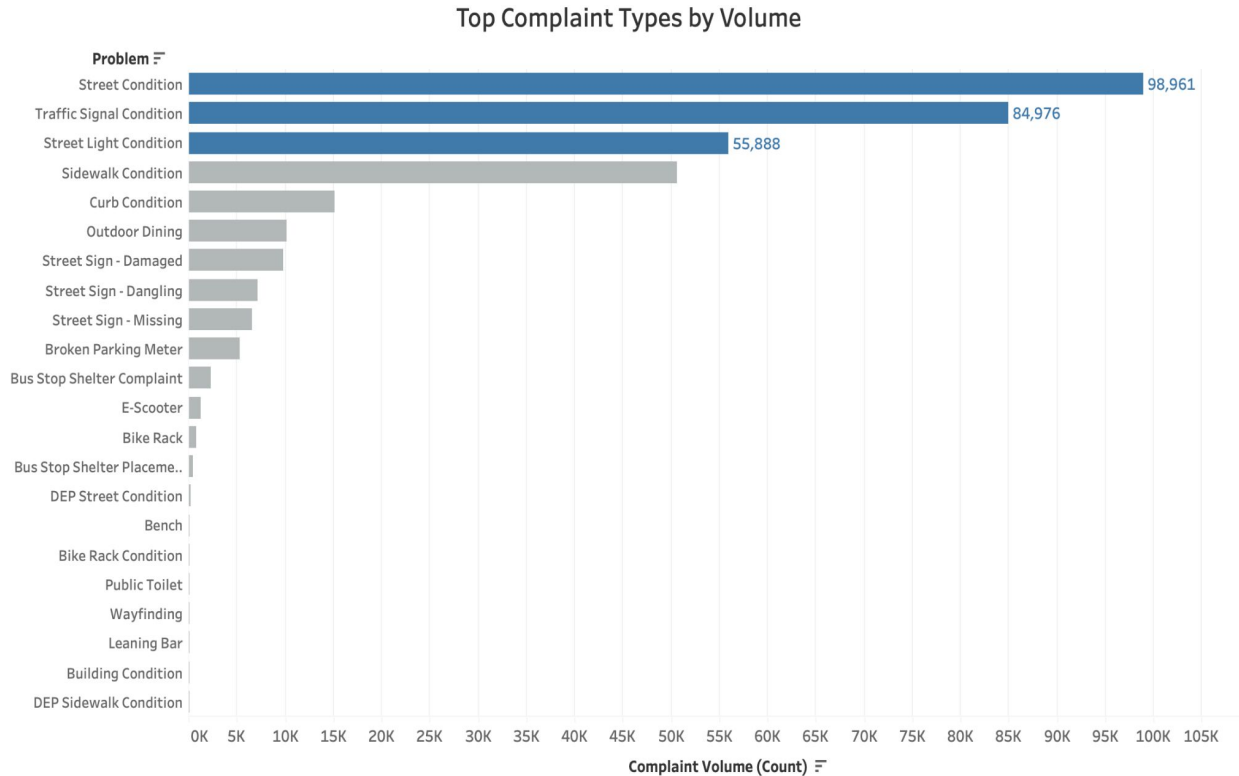
Tools Used

Excel / Google Sheets / Python → Data cleaning & preparation

Tableau → Data visualization & dashboard creation

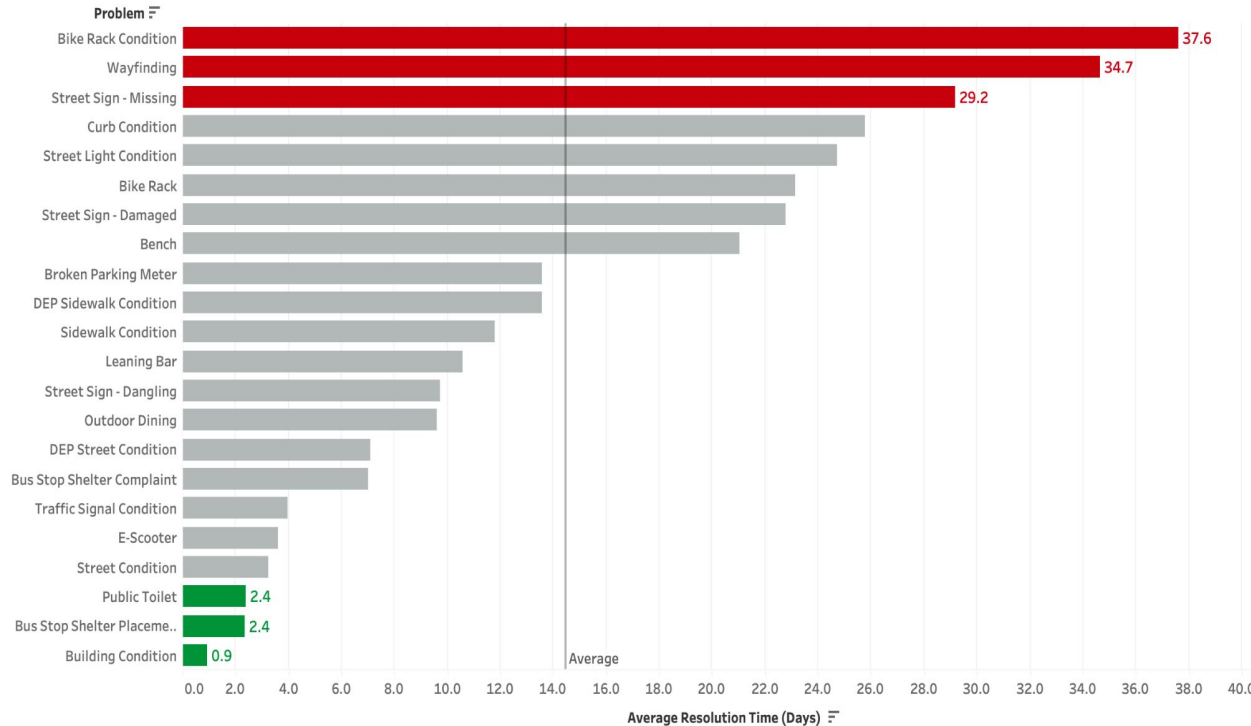
Google Slides → Presentation

Top Complaint Types by Volume



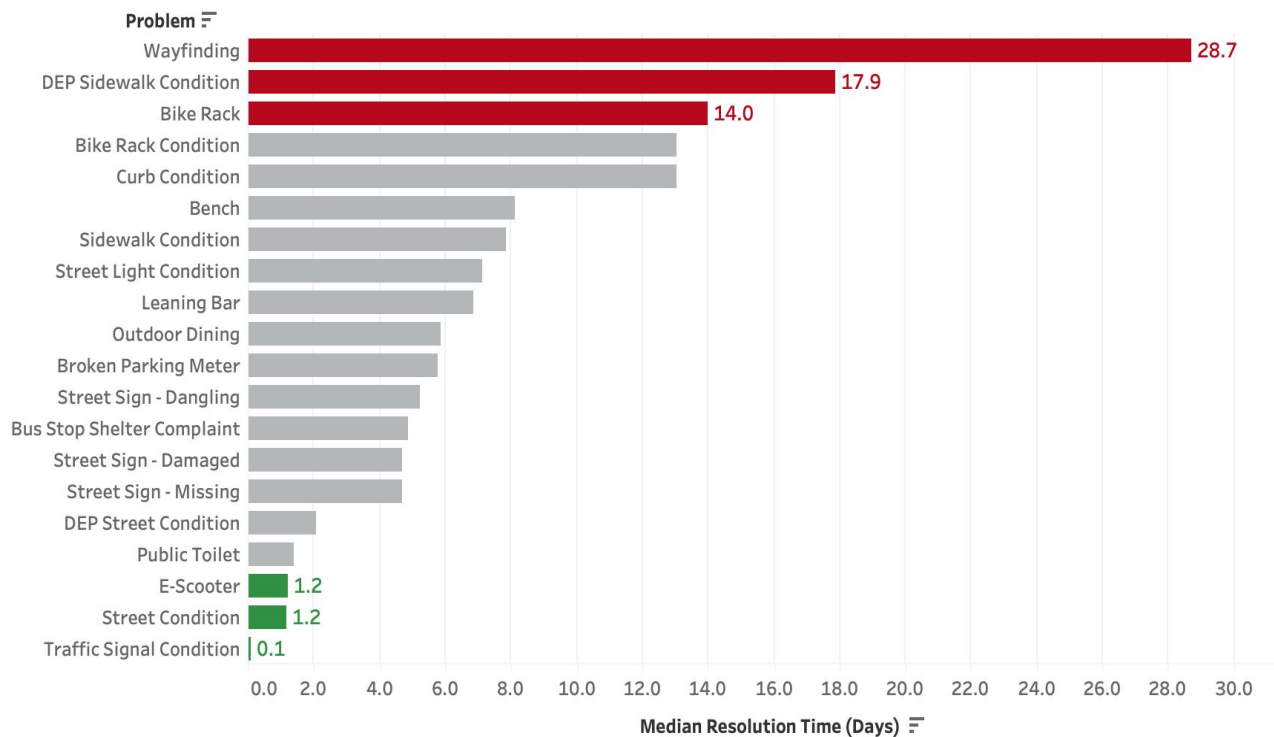
The **top 3** complaint types account for the **majority** of all 311 DOT requests.

Which Complaint Types Take Longer to Resolve on Average?



Certain complaint types take significantly longer to resolve, often exceeding 30 days on average.

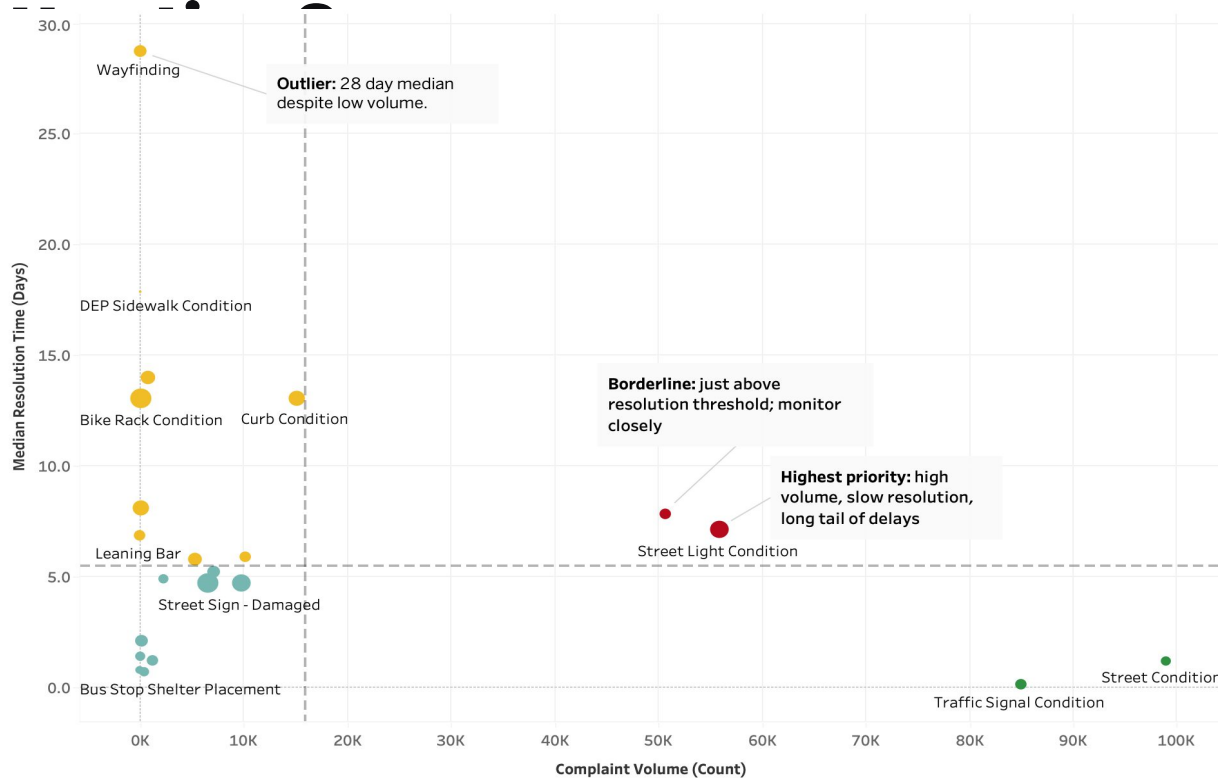
Median Resolution Time Shows Typical Response Speed by Complaint Type



Lower median times compared to **averages** indicate that a **small number of extreme delays** are driving **higher averages**.

Which Complaint Types Need the Most

A

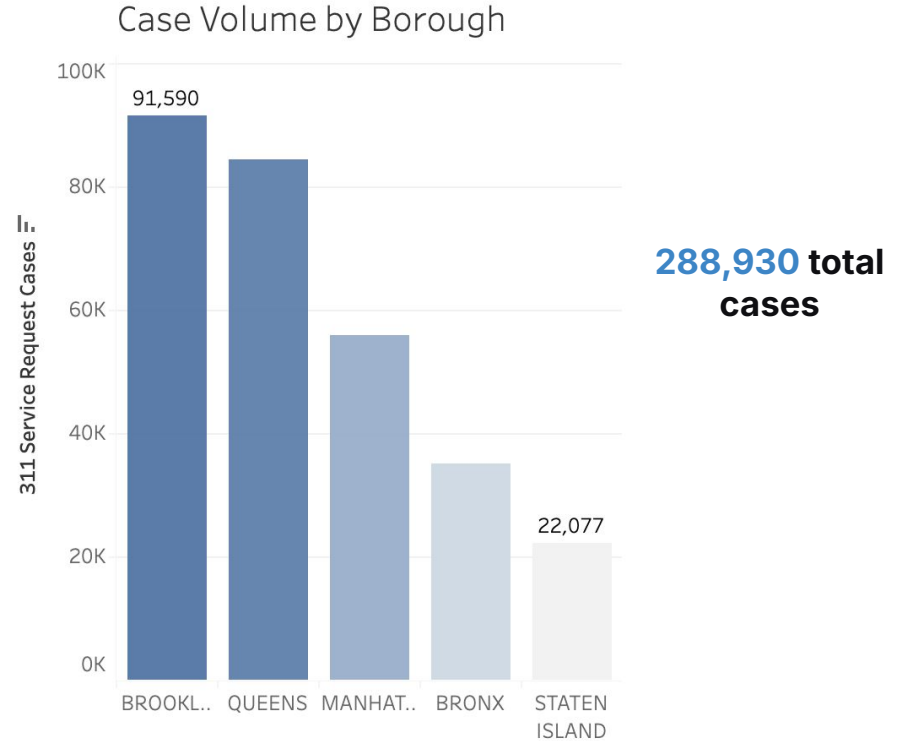


High volume but resolved quickly → **operationally efficient**

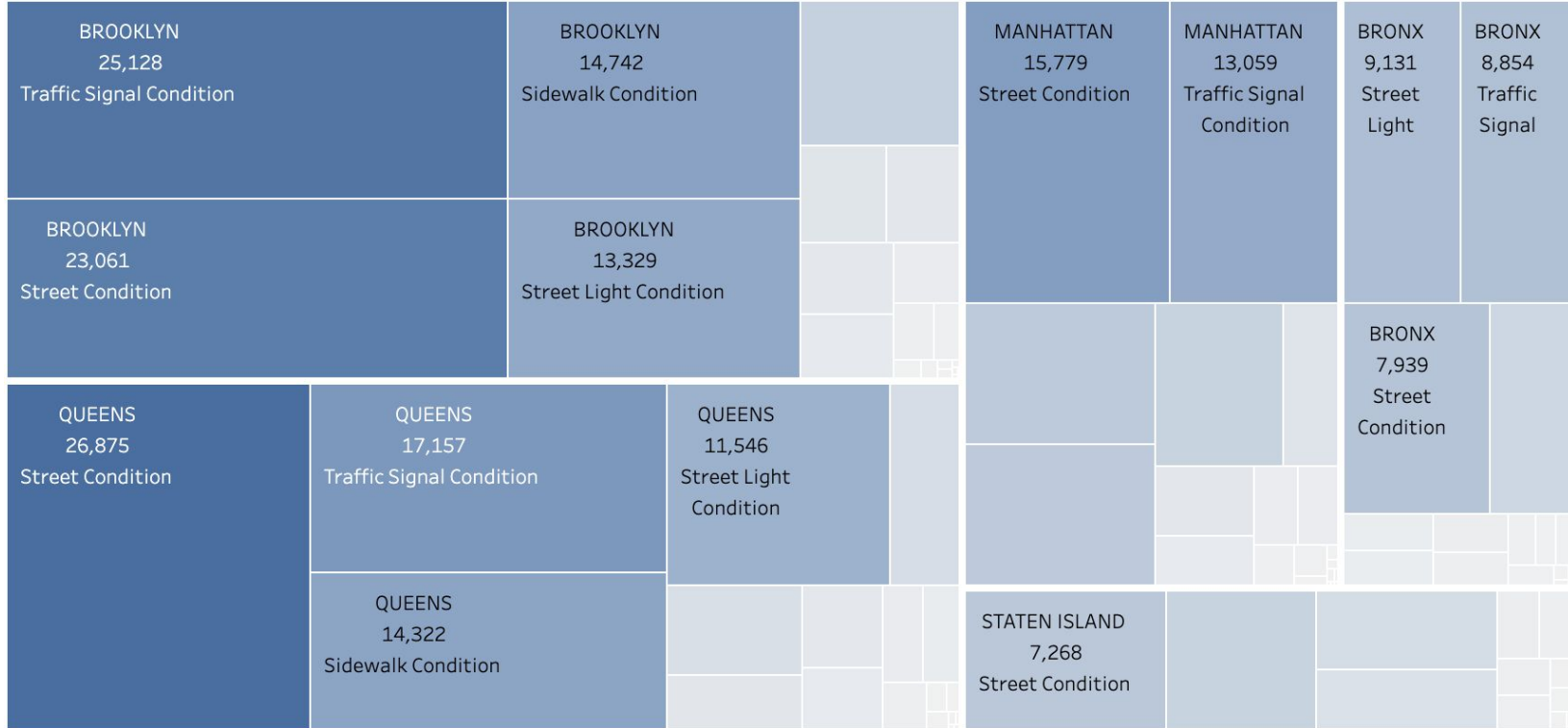
Low volume and fast resolution → **lower priority**

High volume + slow resolution → **top priority for improvement**

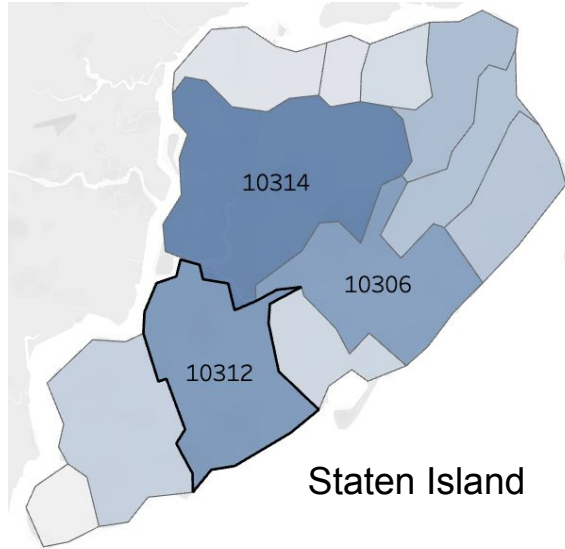
Case Count by Borough



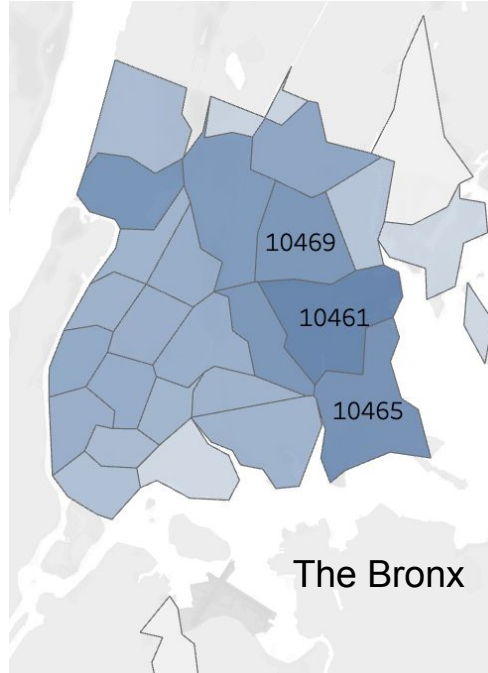
Case Count with Problem by Borough



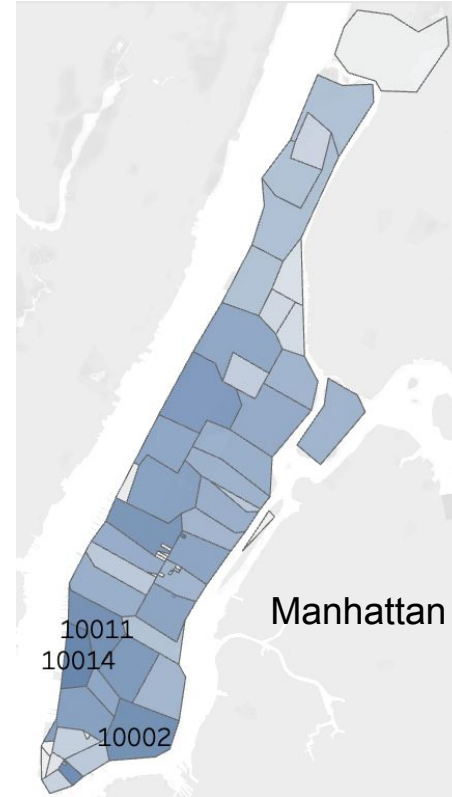
Which zip codes have the most cases?



**10314 with
4,183 cases**

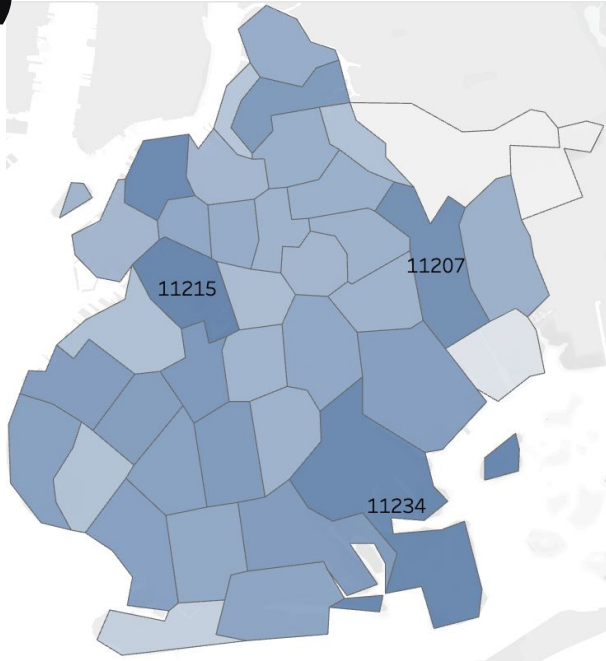


**10461 with
2,460 cases**



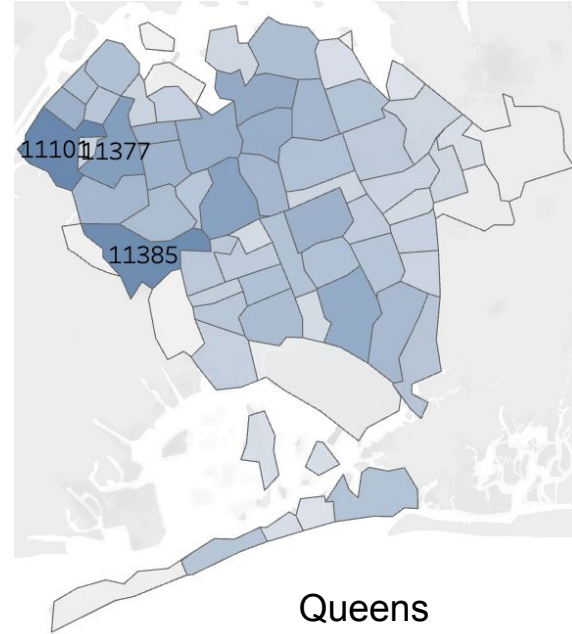
**10014 with
2,728 cases**

Which zip codes have the most cases? (pt. 2)



Brooklyn

11215 with
3,937 cases

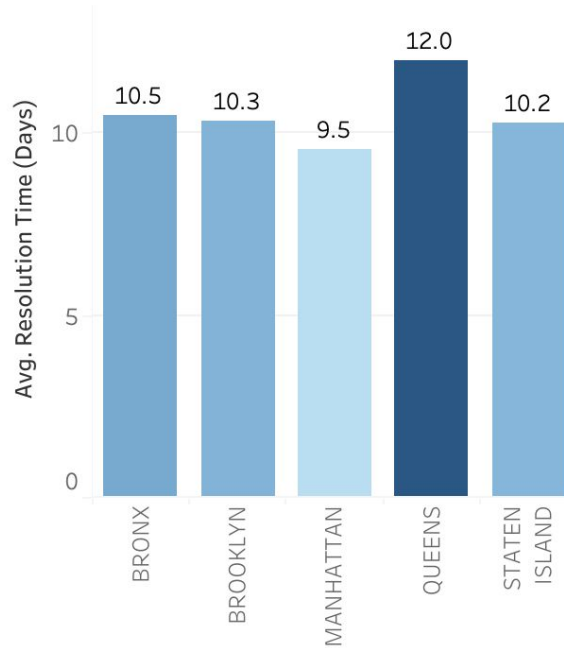


11101 with
3,837 cases

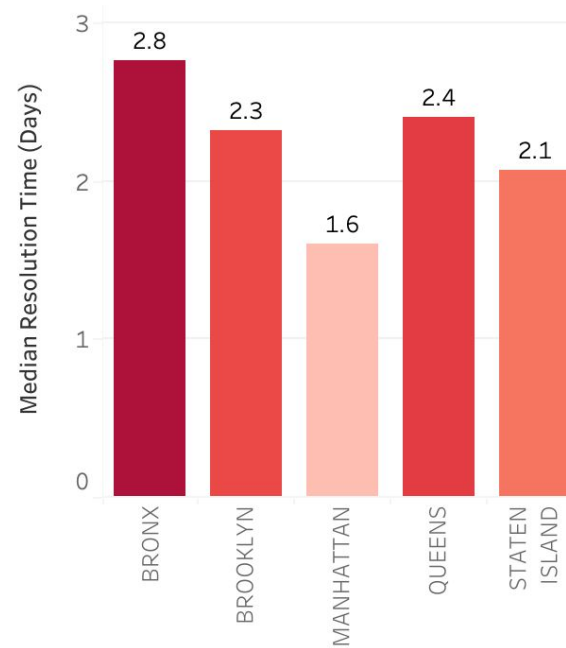
Queens

Average and Median Resolution Time by Borough

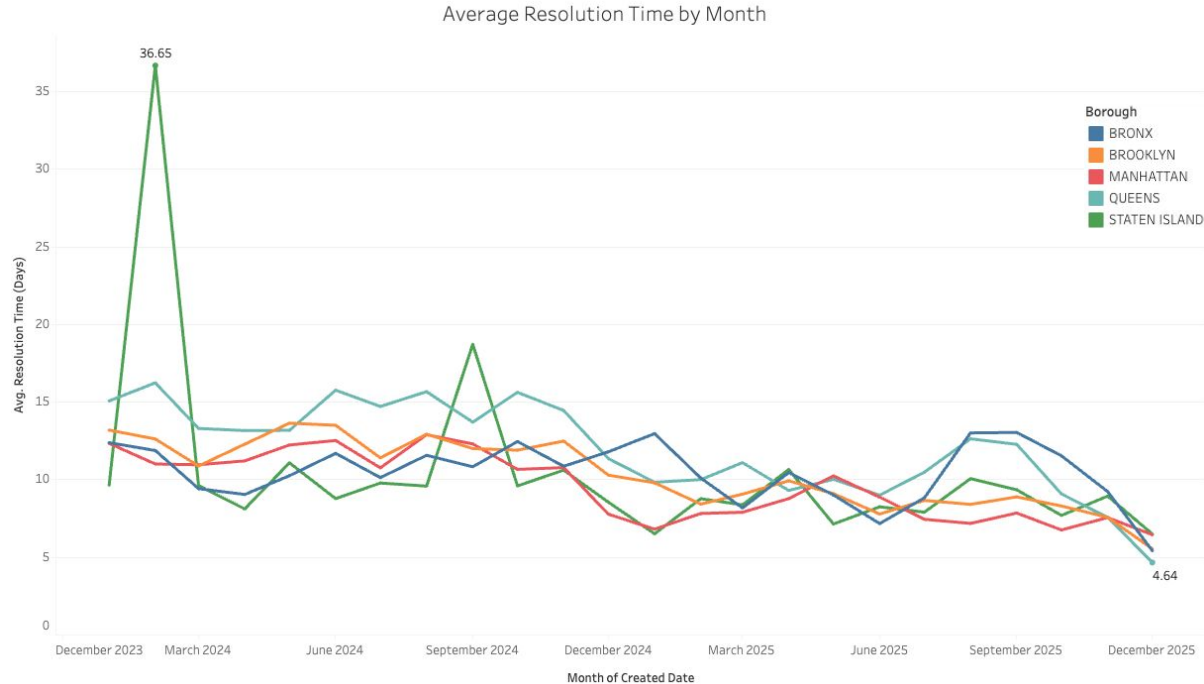
Average Resolution Days



Median Resolution Days



High-volume areas have similar resolution speeds



Queens has **longer average resolution times** compared to other boroughs

Overall, resolution time is **relatively flat** from month-to-month yet trending **downward**

Staten Island observed the **largest spikes** throughout 2024 but has since **normalized**

Resolution times vary widely across problem

Resolution Time by Borough & Problem

Problem	Borough					Grand Total
	BRONX	BROOKLYN	MANHATTAN	QUEENS	STATEN ISLAND	
Bike Rack Condition	26.3	40.7	42.4	26.7	10.1	37.8
Wayfinding	9.5	45.9	36.9	25.7		34.7
Street Sign - Missing	22.5	15.5	24.9	48.7	22.0	29.2
Curb Condition	26.7	23.9	21.2	31.2	26.7	25.8
Street Light Condition	19.3	24.9	30.9	24.7	25.3	24.7
Bike Rack	21.9	22.5	23.8	23.6		23.1
Street Sign - Damaged	19.6	14.5	13.5	35.4	12.4	22.8
Bench	19.0	20.6	28.2	11.7	7.0	21.0
Broken Parking Meter	17.0	15.1	10.8	14.2	20.2	13.6
DEP Sidewalk Condition	15.0	10.0	17.9			13.6
Sidewalk Condition	12.5	11.1	12.0	12.5	10.5	11.8
Leaning Bar	6.8	18.0	8.0		5.9	10.6
Street Sign - Dangling	9.7	8.6	10.6	10.7	8.1	9.7
Outdoor Dining	10.1	9.4	9.5	10.1	14.4	9.6
DEP Street Condition	2.1	7.4	1.4	5.3	101.2	7.1
Bus Stop Shelter Complaint	6.5	7.6	7.0	7.1	5.7	7.0
Traffic Signal Condition	3.0	4.4	1.7	6.0	1.8	4.0
E-Scooter	8.6	2.1	11.4	2.1		3.6
Street Condition	4.2	3.8	2.4	3.2	2.4	3.3
Public Toilet	2.3	2.8	2.2	1.4		2.4
Bus Stop Shelter Placeme..	2.6	1.6	1.9	3.1	3.8	2.4
Building Condition	2.2	0.8	0.5			0.9
Grand Total	10.5	10.3	9.5	12.0	10.2	10.7

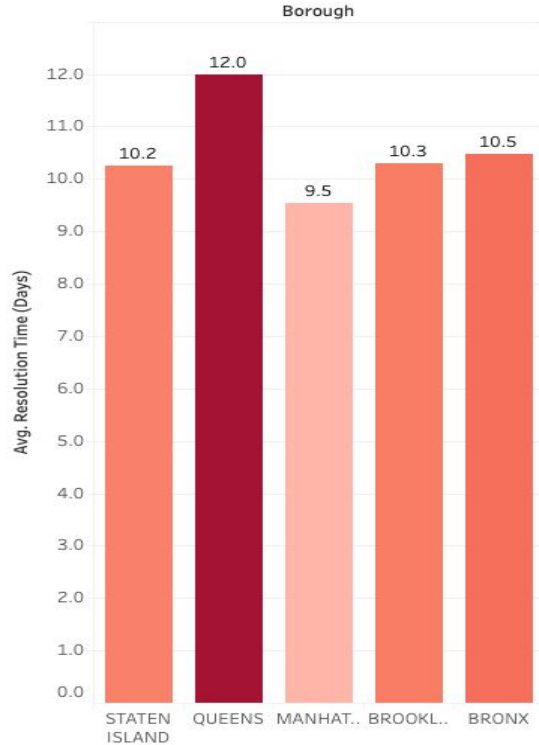
Complaint resolution times are ~10 days on average with a range of 9.5 - 12 days

Queens takes 15 - 30 more days to resolve complaints related to damaged or missing street signs

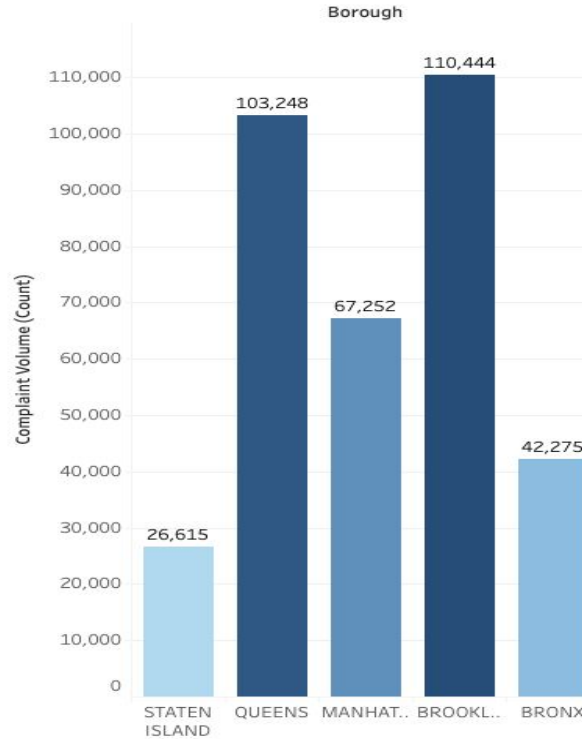
Staten Island takes >100 days to resolve Street Condition complaints, while other boroughs typically take 2 - 7 days

Complaints Volume & Resolution Time

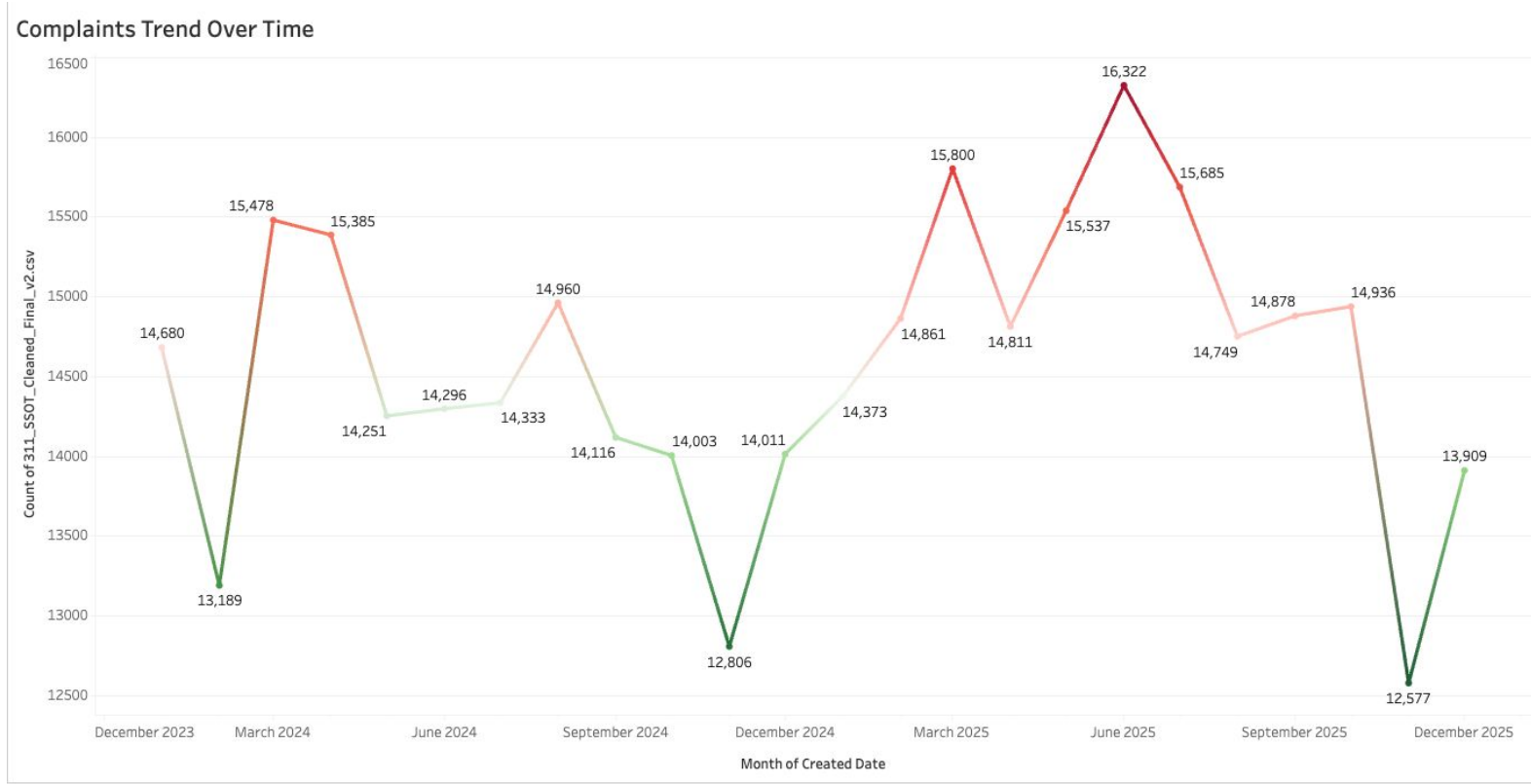
Average Resolution Time



Average Volume per Borough



Complaints Trends





Conclusion

Recap

- **Resolution time** is primarily driven by **complaint type**
- Complaint volume is **concentrated** in a few categories
- Borough-level efficiency differences are **small** but **meaningful**
- High complaint volume **≠** slower response times
- **Extreme outliers significantly impact** average resolution times

Recommendation

- **Allocate additional resources** or specialized teams in Queens and Brooklyn, where both complaint volume and resolution delays are highest
- **Implement predictive workload** balancing across boroughs, using historical complaint volume trends to proactively distribute DOT field teams and reduce backlog accumulation
- **Investigate extreme outliers** in Staten Island resolution times, particularly for Street Condition cases, to identify operational bottlenecks or resource gaps
- **Prioritize infrastructure-related complaints** for process optimization, especially Wayfinding, Missing Street Sign, and Bike Rack issues, which consistently show the longest resolution times

References

Data Source & Tools

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**“Thank you—we're happy
to take any questions.”**